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Project Topic: Sentencing disparities between the sexes

On January 31, 1992; Aileen Wuornos was sentenced to death for the murder of Richard Mallory. The post-trial period of this sentencing consisted of six other murder trials, five of which resulted in a death penalty sentence. These trials were highly publicized and inspired numerous documentaries that make claims on the issues that were highlighted by Aileen's case, including the statement that women convicted of murder are punished more and receive higher sentences than men for similar crimes.

"Are women sentenced more heavily than men for homicide cases?" This question is important as it addresses the potential gender bias in the justice system, which can impact fairness, equality, and public trust. From a societal standpoint, investigating this issue helps ensure that the legal system operates impartially and upholds the principle of justice for all individuals, regardless of gender. If sentencing outcomes differ significantly between men and women for similar crimes, it may point to underlying biases that warrant further examination and policy reform.

From a data analyst's perspective, answering this question requires careful analysis to avoid misleading conclusions due to issues like Simpson's Paradox, where aggregated data (homicide) may hide or reverse trends seen in subgroups (manslaughter and capital murder). I can also use techniques like the Oaxaca-Blinder decomposition to help separate differences in sentencing into explained unexplained components. With my research, my goal is to create a structural model that can automate these parameters and techniques to other types of cases (assault, theft, etc.)

The data I'm using to solve this problem comes from the Criminal Justice Data Transparency (CJDT) framework, which contains many variables for thousands of Florida cases such as gender, race, age, conviction, sentence and use of private/public defender.